

I chose to read the paper “Electrical Resistance of Disordered One-Dimensional Lattices” by Rolf Landauer (1970) because I wanted to understand quantum transport, which is highly applicable in CMOS tech, e.g. Intel’s Clarkdale processor has 383 million transistors, 32 nm channel length. In MOSFETS, when the gate oxide thickness is reduced to less than a few nm, tunnelling between the gate and the channel can become serious problem because well-defined doping profiles can become extremely difficult, so quantum transport seems like an interesting topic to explore. With the wavefunction picture of a particle in mind, given that electrons are quantum waves moving through a 1D chain with random impurities, the paper determines the true electrical resistance when we consider including quantum reflection, interference, and randomness. We have seen the classical Boltzmann transport theory (which is just the classical theory for particle motion and scattering) to some extent previously, so what is new in the quantum picture for this paper is that Landauer treats electrons as quantum waves and calculates how multiple reflections from a disordered sequence of scatterers accumulate. By taking average of an ensemble, the paper finally shows that the effective reflection factor and resistance grows exponentially with the number of scatterers and length of the 1D chain.

First, Landauer discusses how quantum scattering in 1D produces a macroscopic diffusion coefficient D , then he explains how the Einstein relation connects D to resistivity. We imagine a 1D chain with many quantum scatterers (just some object in general). Far left and right are clean regions where wavefunctions are plane waves $e^{\pm ikx}$ as we know from class. Sending electrons from the left produces (details are in the paper):

$$\text{density}_{\text{left}} = 1 + R$$

$$\text{density}_{\text{right}} = 1 - R$$

Over length L , the density gradient is approximately

$$\frac{dn}{dx} \approx -\frac{2R}{L}.$$

Using Fick’s law (the diffusion equation for particles),

$$j = -D \frac{dn}{dx},$$

and the fact that the physical current in the parts with no impurities (at the extremities) is:

$$j = v(1 - R),$$

we obtain:

$$D = \frac{vL(1 - R)}{2R}.$$

If R is small, D is large, so electrons diffuse easily. If R is large, D tends to 0, so diffusion is suppressed. To sum it up, Landauer expresses a macroscopic diffusion coefficient in terms of the quantum scattering property R . This relates transport (for now its D , but we get to resistance later) directly to transmission and reflection of quantum waves through a scattering region. Landauer writes ”An incoherent superposition can be presumed, coherence between waves emanating from the two independent ‘black-body’ reservoirs at the two ends of the chain is not plausible.” Each reservoir is a large piece of metal at some temperature T and chemical potential μ . An electron entering a reservoir undergoes many scattering events, which make random its

phase. Because of this randomisation, the waves originating from the left reservoir and those from the right reservoir do not maintain a fixed phase relationship. And then we can see this is what Landauer calls this an incoherent superposition of left–incident and right–incident waves. When we calculate averaged quantities like density and current, the interference terms (cross terms $\propto \cos \phi$) (I think we did this in class too) disappear. If the phases were not randomised, the whole chain would behave as a perfectly coherent interferometer. The current would then depend on the global interference pattern, we cannot simply say $j = -D \nabla n$, in that sense.

This part requires a bit more background knowledge in thermal physics but for now, I will just assume that we know that at low temperature $k_B T \ll E_F$, only electrons near the Fermi energy (this is just a quantity relevant to Fermi gases) contribute to conduction. So the Einstein relation (which is not derived here as well) becomes:

$$\sigma = e^2 D \left(\frac{\partial n}{\partial \mu} \right)_T, \quad \frac{1}{\rho} = e^2 D \left(\frac{\partial n}{\partial \mu} \right)_T$$

, noting that resistivity is inverse of conductivity. Finally, the macroscopic resistance of length L :

$$\Omega = \rho L = 2 \frac{R}{1-R} \left(\frac{\partial n}{\partial \mu} \right)_T^{-1} e^2 v.$$

The resistance is proportional to $\frac{R}{1-R}$, where R is the quantum reflection probability of the entire 1D scattering region. Real disordered wires do not have perfectly periodic arrangements of obstacles i.e. the distances between impurities are random, so after the resistance result, Landauer now considers an ensemble of chains with randomly spaced obstacles and tries to find for the ensemble average of $\frac{R}{1-R}$ appearing in the resistance, which I noted above. This is Landauer's main ensemble definition:

- Obstacles do not overlap
- The distance between the n th and $(n+1)$ th obstacle is a random variable
- Distances between different pairs are statistically independent
- As a result, the total length L of a particular chain is random also
- The phase shift picked up by an electron traveling between two obstacles is $\phi = kx$, k the wave number at the Fermi energy and x the spacing
- Because the spacings are random, the phases ϕ are taken to be uniformly distributed on 0 to 2π]

Landauer considers two adjacent obstacles with reflection probabilities r_1 and r_2 for each individual obstacle. And he states that when they are combined there are multiple reflected paths which interfere. After some calculations which I have not fully worked through yet, the combined reflection probability r_t obeys

$$\frac{1}{1-r_t} = \frac{1 + r_1 r_2 + 2(r_1 r_2)^{1/2} \cos \phi}{(1-r_1)(1-r_2)}, \quad \frac{r_t}{1-r_t} = \frac{r_1 + r_2 + 2(r_1 r_2)^{1/2} \cos \phi}{(1-r_1)(1-r_2)},$$

where ϕ is an effective phase including both propagation between the obstacles and phase shifts upon reflection.

The rest of the paper involves choosing all obstacles identical, with single-obstacle reflection probability r . Let r_{n-1} be the reflection probability of a chain of $n-1$ obstacles, and r_n that of n obstacles. Taking

$$r_1 = r, r_2 = r_{n-1},$$

and average over the random phase ϕ (so that $\langle \cos \phi \rangle = 0$), we obtain the interesting recursion relations:

$$\frac{1}{1 - r_n} = \frac{1 + rr_{n-1}}{(1 - r)(1 - r_{n-1})}, \quad \frac{r_n}{1 - r_n} = \frac{r + r_{n-1}}{(1 - r)(1 - r_{n-1})}.$$

These are recursion relations for the reflection probability of the n -obstacle chain.

Defining

$$x_n = \frac{1}{1 - r_n}, y_n = \frac{r_n}{1 - r_n},$$

and for a single obstacle

$$x = \frac{1}{1 - r}, y = \frac{r}{1 - r},$$

we can get:

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} = \begin{pmatrix} x & y \\ y & x \end{pmatrix} \begin{pmatrix} x_{n-1} \\ y_{n-1} \end{pmatrix}.$$

After that, Landauer wants $\langle x_n \rangle_{AV}$ and $\langle y_n \rangle_{AV}$, and he does some calculations which I've not fully gone through yet (and he did not really go through it as well). Anyway, the final result for the ensemble average is:

$$\left\langle \frac{R}{1 - R} \right\rangle_{AV} = \langle y_n \rangle_{AV} = \frac{1}{2} \left(\frac{1 + r}{1 - r} \right)^n - \frac{1}{2}.$$

Since the resistance Ω is proportional to $\frac{R}{1 - R}$, we have the fundamental result:

$$\Omega \propto \left(\frac{1 + r}{1 - r} \right)^n,$$

so the resistance of a 1D disordered chain grows exponentially with length and number of obstacles. At least from what I read online, this paper's ideas anticipated the modern Landauer–Buttiker picture (not sure about the details), and form foundation for later results such as the Landauer conductance formula and mesoscopic transport, which make up the field of quantum transport. So this has been a pretty interesting introductory foundational paper.